



Module 11 Day 3: Cryptography

Solution Guide: Cryptography Refresher

The goal of this activity was to practice using digital signatures with GPG, and to demonstrate that digital signatures can be used to send messages that are guaranteed to be authentic.

This solution is an example, as you will each have a different key and message. The following example demonstrates how to sign and verify a message with GPG.

- First, write a message to a file using echo.
 - Nano or Vi can be used to create a message. The echo command is:
 - `echo "This is an Urgent reminder by the security team of Hill Valley PD. Please be sure to use GPG to sign and verify all messages" > Important_Communication_by_me.txt`
 - Be sure to replace the word "me" with your name.
- Sign the message with your private key and create an output file called `Important_Communication_by_me.sig`:
 - `gpg --output Important_Communication_by_me.sig --clearsign Important_Communication_by_me.txt`
- Run the script and display the clearsigned signature by running the following command:
 - `cat Important_Communication_by_me.sig`
 - This should display the following:

Unset

-----BEGIN PGP SIGNED MESSAGE-----

Hash: SHA512

This is an Urgent reminder by the security team of Hill Valley PD.

Please be sure to use GPG to sign and verify all messages

-----BEGIN PGP SIGNATURE-----

iQGzBAEBCgAdFiEEYS1ws2SzZx5pJqjYHX0Iu8WCN4gFA140Bv8ACgkQHX0
Iu8WC

N4gf/gv+PPq/qZhETpk1zIjRpn620H2LpUIPqIEwqyoZ8A7WuLMTyJNjUpp
6Q43T

ZnL8fcBj5c51AswFE1I4tGbWASc/YUISZCv/hMZ2q126rbTZnfvVWG1IHEu
Nw1La

01e3C7iRB845nFuY3YZG1Kdr2a0X+M/SPEgv0nQP0MIYIfmCILz20ZxJjyn
9H47d

lgAZiijoMMn10vX60BcLxj5aov6K2YgMDw63z9Z/CB/+Ue2t+M1rm/A9zAQ
xRiNS

V3pdHHaBiDx3s3fieWmPR0/HJK8zU0pkzPs/SLZUkz0WZqvLv6RHE9Viab4
YwjIP

ZCaMjSrkcVUq9KwbpWE6ZFdVT3Cq/dNgRRxTuCImoRA0XzLEwLpBQIbFBoz
lGM/p

ZDdzH0YkK5ZlnAF/rtmQ0CVE5GFyipTXEsD2Sg4Tj52AdT14JLKjRff5U1W
sBW8D

qJgSybjnY0GaABMIVCTAoUMC4dxGoEbZxPZZFfZKoxNViTykFIuw3xvzG0v
TYz5h

HEqp1TYb

=skQM

-----END PGP SIGNATURE-----

- This shows the plaintext message and the separate digital signature in the same file.
- We will pretend that the message above is the one we received from our partner.
 - Since you should already have your partner's public key imported from the previous activity, we don't have to repeat that step.
- Verify the message by running the following command:
 - `gpg --verify Important_Communication_by_me.sig`
- The verification response should look similar to the following:

Unset

```
gpg: Signature made Mon 09 Dec 2019 02:30:52 PM EST
gpg:                using RSA key
39B2BD6C93E1E63E8C004183FE91AF7A7B4EC267
gpg: Good signature from "me <metest.com>" [ultimate]
```